RAJALAKSHMI ENGINEERING COLLEGE

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D2

Time left 0:40:01

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdlib.h>

// Binary search to find the upper bound (first element > key)
int upper_bound(int arr[], int size, int key) {
    int low = 0, high = size;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= key) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int lengthOfLNDS(int nums[], int n) {
    if (n <= 0) return 0;

    int *tails = (int *)malloc(n * sizeof(int));
    int length = 0;

    for (int i = 0; i < n; i++) {
        int x = nums[i];
        int idx = upper_bound(tails, length, x);

        tails[idx] = x;
        if (idx == length) {
            length++;
        }
    }

    free(tails);
    return length;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int *nums = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    printf("%d\n", lengthOfLNDS(nums, n));

    free(nums);
    return 0;
}
```

Check

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔

	Input	Expected	Got	
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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